WORKED SOLUTION

STEP II 1997, Question 9

Mechanics • Friction and Rigid-Body Equilibrium

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Read the question before you go on. It is in the STEP Support Programme's question database:

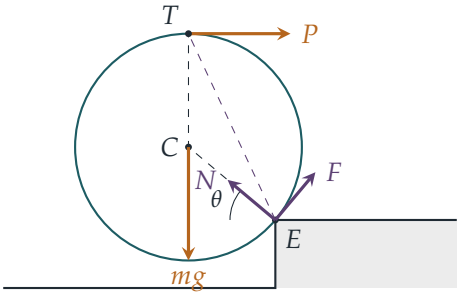
step.maths.org/questions/97-s2-q9

The key idea

Because the sphere is drawn *slowly*, it is in equilibrium at every instant, so in each position we have a statics problem. Once the sphere has left the ground only three forces act on it: the weight, the pull at the top and the reaction at the edge of the step. Taking moments about the edge gets rid of the unknown reaction and gives the pull. The three-force rule (the three lines of action must meet at a point) then tells us exactly which way the reaction points, and that gives the friction condition almost for free.

Solution

Let the radius be $r = \frac{1}{2}d$, so the step has height $\frac{1}{4}d = \frac{1}{2}r$. We model the sphere as a rigid uniform body touching the step only at the edge E (a horizontal line, which in the vertical cross-section through the centre is a point). Let C be the centre and T the highest point, where the horizontal force P acts towards the step. Let θ be the angle that EC makes with the horizontal.



The geometry

With E as origin, C is at height $r\sin\theta$ above E and a horizontal distance $r\cos\theta$ from it, and T is a further r above C. At the start, when the sphere rests on the ground, C is at

height r above the ground, so $\frac{1}{2}r$ above E; hence $\sin\theta = \frac{1}{2}$ and $\theta = \frac{\pi}{6}$. The sphere is fully on the step when C is vertically above E, i.e. $\theta = \frac{\pi}{2}$. So the movement corresponds to θ increasing from $\frac{\pi}{6}$ to $\frac{\pi}{2}$.

The maximum force

While the sphere still touches the ground, the ground carries part of the weight, and it can only push upwards. The sphere starts to rise when P is just large enough for the ground reaction to fall to zero. From then on the forces are P at T, the weight mg at C, and the reaction of the edge at E, with normal component N along EC and friction F perpendicular to EC.

Taking moments about E (the reaction has no moment there), the arm of P is the height of T above E, which is $r(1+\sin\theta)$, and the arm of the weight is $r\cos\theta$. Equilibrium gives

$$P\,r(1+\sin\theta) = mg\,r\cos\theta \quad\Longrightarrow\quad P = \frac{mg\cos\theta}{1+\sin\theta}.$$

On $\frac{\pi}{6} \le \theta \le \frac{\pi}{2}$ the numerator decreases and the denominator increases, so P is decreasing in θ. (Before lift-off, P only builds up to this value.) The maximum is therefore at $\theta = \frac{\pi}{6}$, the moment of leaving the ground:

$$P_{\max} = \frac{mg\cdot\frac{\sqrt{3}}{2}}{1+\frac{1}{2}} = \frac{mg}{\sqrt{3}}.$$

The friction condition

Ask yourself Which way does the reaction at the edge point?

Three non-parallel forces in equilibrium have concurrent lines of action. The line of P is the horizontal through T and the line of the weight is the vertical through C, which also passes through T. So they meet at T, and the total reaction at E must act along ET.

Now $CE = CT = r$, so triangle CET is isosceles. The angle at C between CT (vertically up) and CE (pointing down at angle θ below the horizontal) is $\frac{\pi}{2}+\theta$, so each base angle is

$$\angle CET = \tfrac{1}{2}\left(\pi - \tfrac{\pi}{2} - \theta\right) = \tfrac{\pi}{4} - \tfrac{\theta}{2}.$$

This is the angle between the total reaction and the normal EC, so

$$\frac{F}{N} = \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right).$$

We can confirm this by resolving. The reaction is $(-P,\ mg)$, since it balances P and the weight. Its component along the unit normal $(-\cos\theta, \sin\theta)$ and along the unit tangent

$(\sin\theta, \cos\theta)$ are

$$N = P\cos\theta + mg\sin\theta = mg\,\frac{\cos^2\theta + \sin\theta + \sin^2\theta}{1+\sin\theta} = mg,$$
$$F = mg\cos\theta - P\sin\theta = mg\cos\theta\left(1 - \frac{\sin\theta}{1+\sin\theta}\right) = \frac{mg\cos\theta}{1+\sin\theta},$$

so $F/N = \cos\theta/(1+\sin\theta)$, which equals $\tan(\frac{\pi}{4} - \frac{\theta}{2})$ by the half-angle formulae. Both F and N are positive throughout, so the edge really does push on the sphere and friction acts up the tangent, as drawn.

There is no slipping at E only if $F \le \mu N$ in every position, i.e.

$$\mu \ge \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \quad \text{for all } \tfrac{\pi}{6} \le \theta \le \tfrac{\pi}{2}.$$

The right-hand side decreases as θ increases, so the most demanding position is again $\theta = \frac{\pi}{6}$, where it equals $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$. At that instant friction must be at least $1/\sqrt{3}$ of the normal reaction; with $\mu = 1/\sqrt{3}$ exactly, friction would be limiting and the sphere on the point of slipping, so to draw it up safely without slipping we need $\mu > 1/\sqrt{3}$, as required.

Checks

- At $\theta = \frac{\pi}{2}$ the centre is directly above the edge: $P = 0$ and $F = 0$, as they should be.
- $N = mg$ throughout, which makes sense: the reaction is $(-P, mg)$, and its component along the normal works out to exactly mg.
- Numerically, stepping θ from $\frac{\pi}{6}$ to $\frac{\pi}{2}$ with moments and resolution done separately, the largest value of P/mg and of F/N is $0.57735\ldots = 1/\sqrt{3}$, both at the start.

Common slips

- Taking the moment arm of P about E as $r\sin\theta$ (the height of C) instead of $r(1+\sin\theta)$ (the height of T).
- Assuming the reaction at the edge is purely normal. It cannot be: the normal passes through C, and the three forces would then fail to meet at a point.

Beyond STEP

The three-force argument is a special case of a general fact: a rigid body is in equilibrium exactly when the total force and the total moment about any one point are both zero, and for three forces this forces concurrency. The angle between the reaction and the normal, here $\frac{\pi}{4} - \frac{\theta}{2}$, is the "angle of friction" idea, which turns friction problems into geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body statics and dynamics from these principles.

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